

EECE 202– Linear Circuit Analysis II
Spring 2026
Project Deliverable 2-A: Simulation Report

Project Title: Industrial Load Analysis and Power Factor Improvement

Section Number: M4

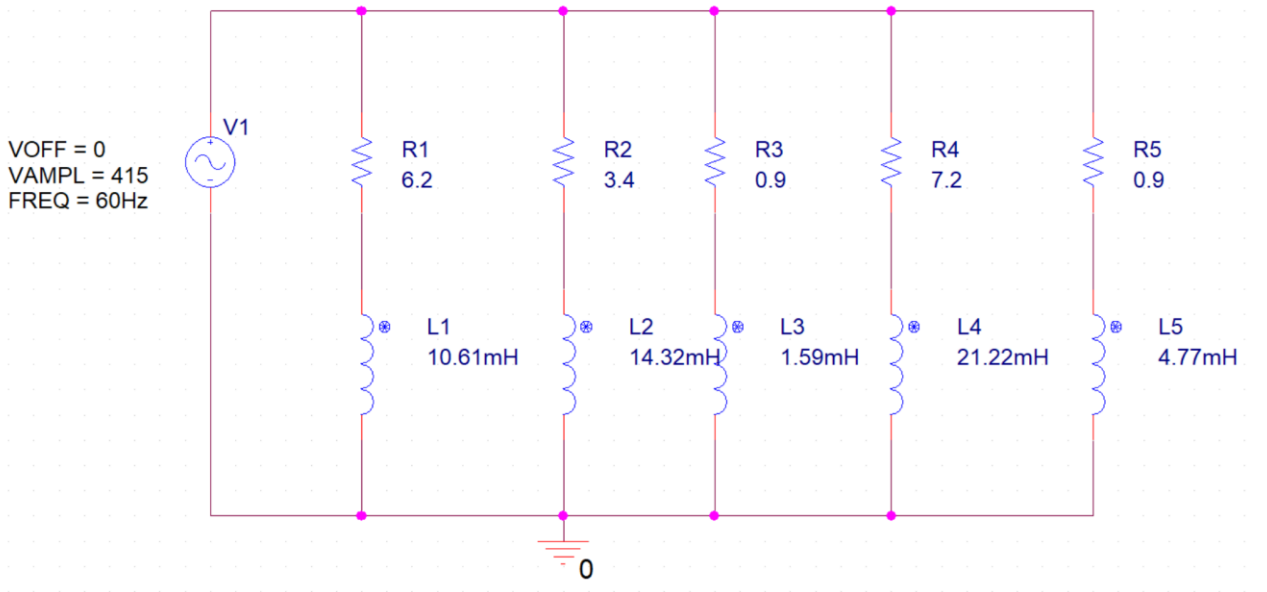
Group Number: Group 4

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1. PSpice Modelling: Original System

1.1. Circuit of the Original System (PSpice Screenshot) – 20%

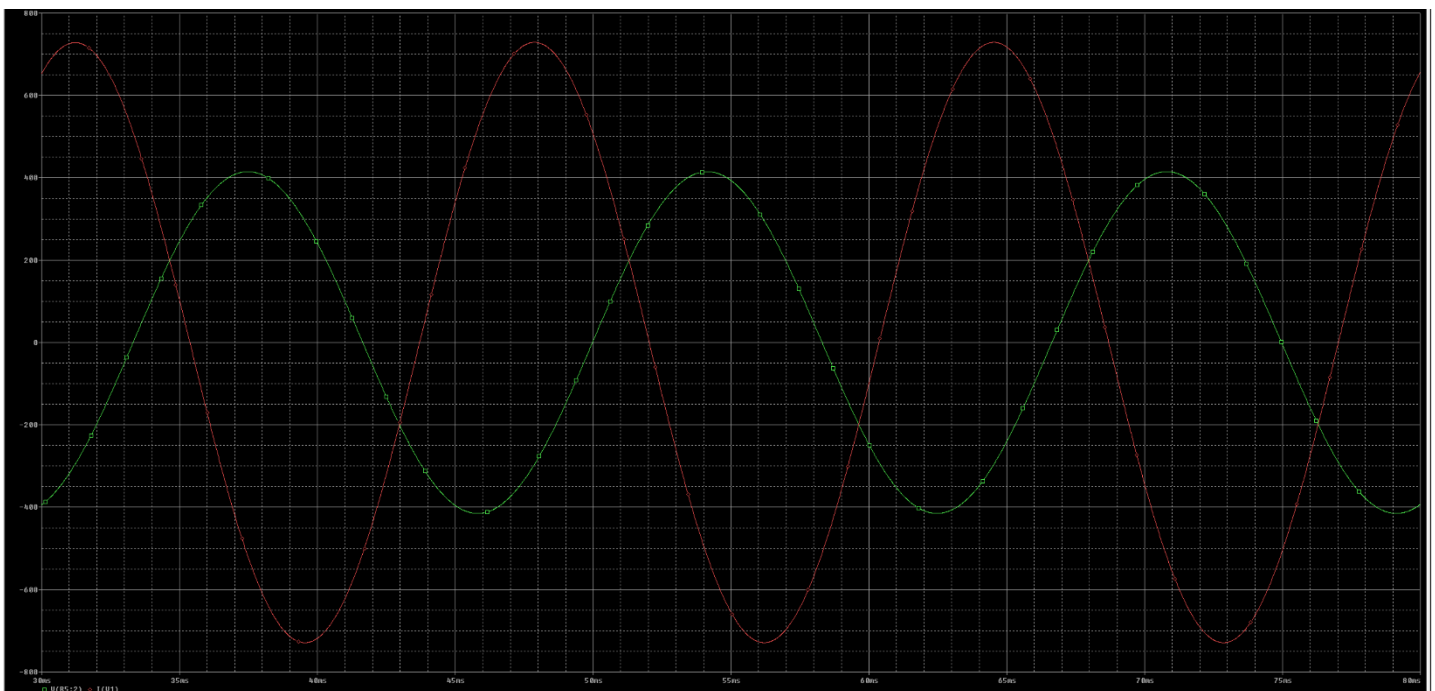
The original industrial load circuit was modelled in OrCAD Capture using a VSIN voltage source (VAMPL = 415 V, FREQ = 60.03 Hz, VOFF = 0) connected to five parallel R-L branches representing loads IL₁ through IL₅. Each load is modelled as a series resistor and inductor with values derived from the theoretical impedances at $\omega = 377$ rad/s.



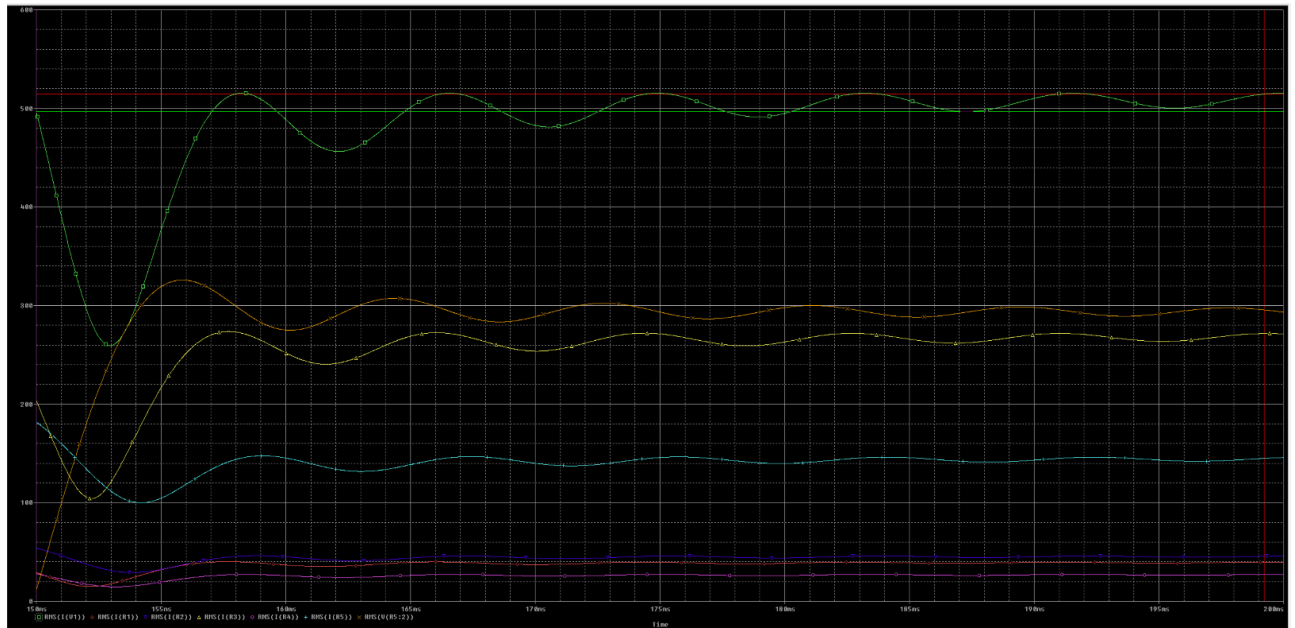
2. Original System Simulation Results – 40%

2.1. Simulation Results

The transient simulation was run for 200 ms with data saved after 150 ms to ensure steady-state conditions. RMS values were obtained using the RMS() trace expression in PSpice



Probe. The source voltage and source current waveforms are shown below, confirming the expected phase lag.



Trace Colo	Trace Nam	Y1
	X Values	199.228
CURSOR 1	RMS(I(V1))	514.560
	RMS(I(R1))	39.825
	RMS(I(R2))	45.760
	RMS(I(R3))	271.674
	RMS(I(R4))	27.190
	RMS(I(R5))	145.022
	RMS(V(R5:2))	295.609

The simulated RMS branch currents and power quantities for the original system are summarized in the tables below.

Load	Type	I_{sim} (A) — Simulated
I_{L1}	Induction Motor	39.83
I_{L2}	Welding Machine	45.76
I_{L3}	Industrial Transformer	271.7
I_{L4}	Conveyor System Drive	27.19
I_{L5}	Air Compressor	145.0
I_{source}	—	514.6
V_{RMS}	—	295.6 V

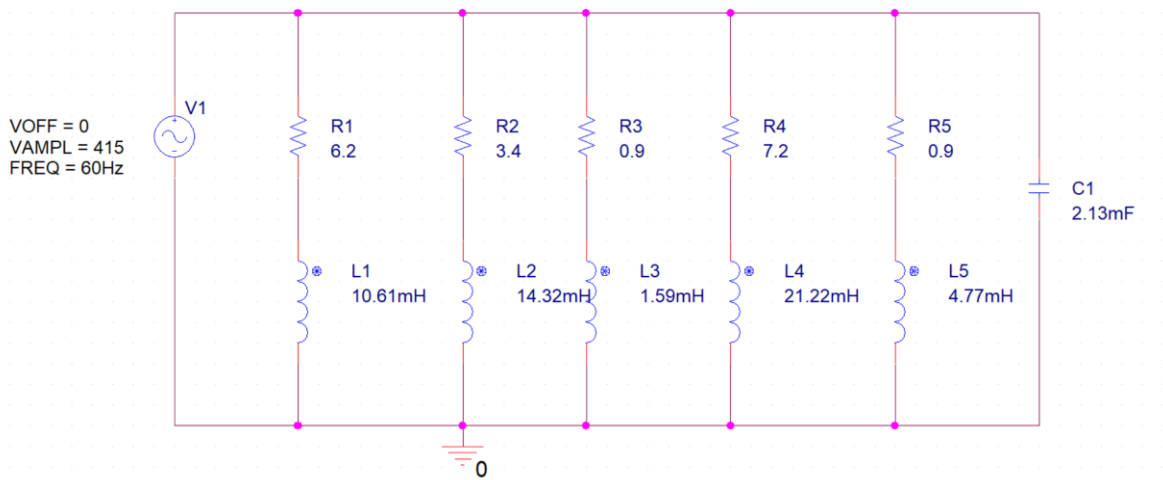
Load	$P_{sim}(W)$	$Q_{sim}(VAR)$	$S_{sim}(VA)$	PF_{sim}
I_{L1}	9,836	6,345	11,717	0.840 lag
I_{L2}	7,122	11,310	13,331	0.534 lag

I_{L3}	66,454	44,303	79,771	0.833 lag
I_{L4}	5,322	5,913	7,930	0.671 lag
I_{L5}	18,923	37,845	42,380	0.447 lag
Total	107,657	105,716	150,566	0.715 lag

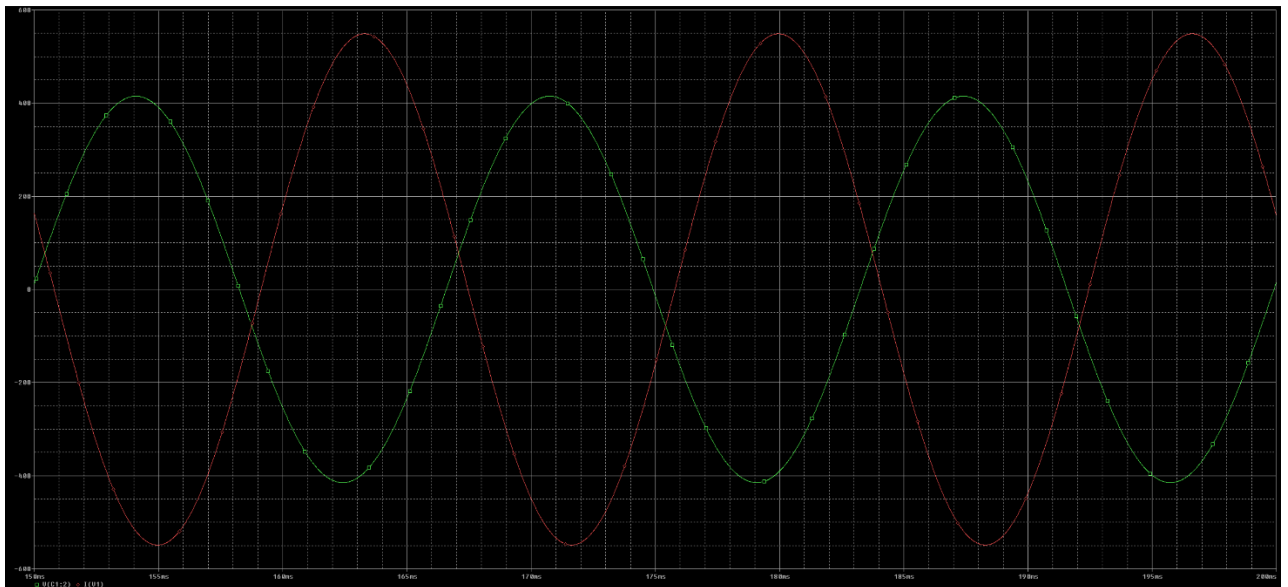
3. Compensated System

3.1. Circuit of the Compensated System (PSpice Screenshot) – 10%

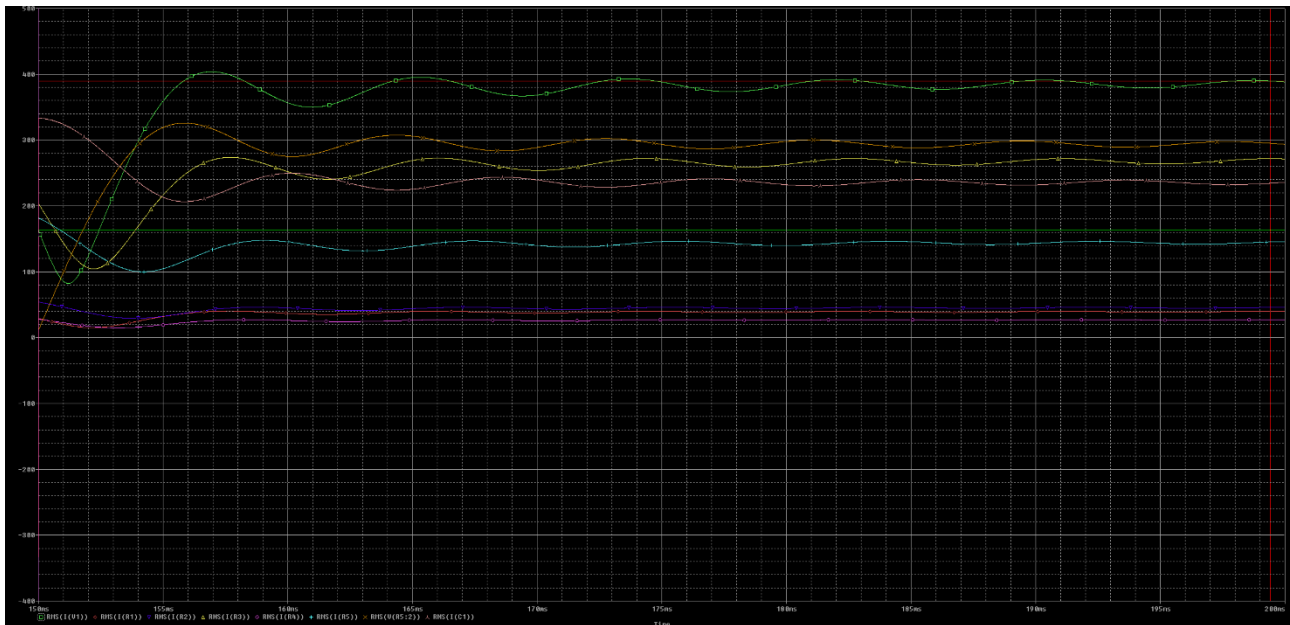
The compensation capacitor $C_1 = 2.18 \text{ mF}$ was added as a sixth parallel branch to the original circuit. This value was derived theoretically to supply $Q^c = 70.7 \text{ kVAR}$ of leading reactive power, raising the system power factor from 0.713 to a target of 0.950 lagging.



3.2. Simulation Results – 10%



The compensated circuit was simulated under identical conditions. The source current waveform shows a visibly reduced phase lag compared to the original circuit, confirming the improvement in power factor.



Trace Color	Trace Name	Y1
	X Values	199.418m
CURSOR 1,2	RMS(I(V1))	389.665
	RMS(I(R1))	39.826
	RMS(I(R2))	45.833
	RMS(I(R3))	271.697
	RMS(I(R4))	27.219
	RMS(I(R5))	145.287
	RMS(V(R5:2))	295.067
	RMS(I(C1))	234.448

The simulated RMS values for the compensated system are:

Load	Type	I_{sim} (A) — Simulated
I_{L1}	Induction Motor	39.83 (unchanged)
I_{L2}	Welding Machine	45.83 (unchanged)
I_{L3}	Industrial Transformer	271.7 (unchanged)
I_{L4}	Conveyor System Drive	27.22 (unchanged)
I_{L5}	Air Compressor	145.3 (unchanged)
I_{C1}	Compensation Capacitor	234.4
I_{source}	—	389.7
V_{RMS}	—	295.1 V

Quantity	Calculation	Value
P_{total}	9,836+7,142+66,454+5,330+19,003	107,765 W $\approx$ 107.8 kW
Q_{total}	6,346+11,352+44,303+5,923+38,006	105,930 VAR
Q_C	$I_{C1}^2 \times X_C = 234.4^2 \times 1.218$	66,979 VAR $\approx$ 67.0 kVAR

Q_{net}	105,930 – 66,979	38,951 VAR $\approx$ 38.95 kVAR
S_{total}	$\sqrt{107,765^2 + 38,951^2}$	114,641 VA $\approx$ 114.6 kVA
PF	107,765 / 114,641	0.940 lagging

4. Comparison Between Theoretical and Simulation Results – Original System – 5%

The table below compares theoretical values from PD1 with the PSpice simulation results for the original (uncompensated) system.

Parameter	Theoretical	Simulated	Error (%)
V_{RMS} (V)	293.4	295.6	0.75%
I_{source} (A)	515.0	514.6	0.08%
$I_{R1} - I_{L1}$	39.77	39.83	0.15%
$I_{R2} - I_{L2}$	45.97	45.76	0.46%
$I_{R3} - I_{L3}$	271.2	271.7	0.18%
$I_{R4} - I_{L4}$	27.26	27.19	0.26%
$I_{R5} - I_{L5}$	145.8	145.0	0.55%
P_{total} (kW)	107.7	107.7	0.00%
Q_{total} (kVAR)	106.1	105.7	0.38%
S_{total} (kVA)	151.1	150.6	0.33%
PF	0.713	0.715	0.28%

5. Comparison Between Theoretical and Simulation Results – Compensated System – 5%

The table below compares theoretical values from PD1 with the PSpice simulation results for the compensated system (with $C_1 = 2.18$ mF).

Parameter	Theoretical	Simulated	Error (%)
V_{RMS} (V)	293.4	295.1	0.58%
I_{source} (A)	386.5	389.7	0.83%
$I_{R1} - I_{L1}$ (A)	39.77	39.83	0.15%
$I_{R2} - I_{L2}$ (A)	45.97	45.83	0.30%
$I_{R3} - I_{L3}$ (A)	271.2	271.7	0.18%
$I_{R4} - I_{L4}$ (A)	27.26	27.22	0.15%

$I_{R5} - I_{L5}(\text{A})$	145.8	145.3	0.34%
$I_{C1}(\text{A})$	240.9	234.4	2.70%
$P_{total}(\text{kW})$	107.7	107.8	0.09%
$Q_{net}(\text{kVAR})$	35.4	38.95	10.03%
$S_{total}(\text{kVA})$	113.4	114.6	1.06%
PF	0.950	0.940	1.05%

6. Discussion of Discrepancies – 10%

The simulation results for the original system show excellent agreement with the theoretical calculations, with all branch current errors below 0.6% and overall system power factor error of only 0.28%. These minor discrepancies are attributable to the following factors:

- **Frequency approximation:** The theoretical analysis uses $\omega = 377 \text{ rad/s}$ exactly, corresponding to $f = 59.97 \text{ Hz}$. The simulation uses $FREQ = 60.03 \text{ Hz}$, introducing a small frequency deviation that slightly affects inductive reactances ($X_L = \omega L$), and consequently the branch currents.
- **RMS convergence:** The *RMS* function in PSpice computes a running time-average from $t = 0$. Even with data saved after 150 ms , the traces exhibit residual oscillation at 200 ms . This causes slight reading errors when the cursor is placed at a non-peak point of the *RMS* ripple.
- **Ideal component modelling:** PSpice treats all the resistors and inductors as ideal elements with no parasitic resistance, capacitance, or frequency dependence. In practice, real inductors have winding resistance and core losses that would slightly alter impedance values.

The compensated system shows slightly larger discrepancies, particularly in Q_{net} (10.03% error) and PF (1.05% error). The primary cause is the underestimation of $I(C_1)$: the simulated capacitor current of 234.4 A is 2.70% lower than the theoretical 240.9 A . This occurs because:

- The *RMS* () trace for $I(C_1)$ has not fully converged at 200 ms due to the large capacitance value (2.18 mF), which has a long charging transient at startup that biases the running average downward.
- As a result, the reactive power cancelled by the capacitor is underestimated at 67.0 kVAR (simulated) vs 70.7 kVAR (theoretical), leaving more net reactive power $Q_{net} = 38.95 \text{ kVAR}$ in the simulation vs 35.4 kVAR theoretically.
- Running the simulation to a longer time window (e.g. 500 ms) would allow $I(C_1)$ to converge fully and bring the simulated PF closer to the theoretical 0.950.

Despite these discrepancies, all simulation results are within acceptable engineering tolerances (under 3% for currents, under 11% for reactive power). The simulation successfully validates the theoretical design: the compensated system power factor of 0.940 clearly demonstrates a significant improvement over the original 0.715, confirming that the shunt capacitor bank design is effective and the theoretical calculations are correct.